WORKED SOLUTION

STEP II 1988, Question 8

Pure • Matrices and Eigenvectors

STEP II 1988, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q8

The key idea

One year of the model is a linear map, so it is multiplication by a 2×2 matrix M. A population whose ratio never changes is one that M just rescales: an *eigenvector*. Once we have the two eigenvectors, any starting population splits into a piece along each, and each piece simply gets multiplied by its own eigenvalue every year. The long-term fate is then decided by which eigenvalue is larger and by the sign of the coefficient in front of it.

Solution

Ratios that stay the same

Writing the populations as a column vector, one year of the model is

$$\begin{pmatrix} A' \\ B' \end{pmatrix} = M \begin{pmatrix} A \\ B \end{pmatrix}, \qquad M = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

The ratio $A : B$ is unchanged exactly when $(A', B') = \lambda(A, B)$ for some number λ, that is, when (A, B) is an eigenvector of M. The eigenvalues satisfy

$$\det(M - \lambda I) = \left(\tfrac{1}{4} - \lambda\right)\left(\tfrac{3}{2} - \lambda\right) + \tfrac{3}{8} = \lambda^2 - \tfrac{7}{4}\lambda + \tfrac{3}{4} = 0,$$

so $4\lambda^2 - 7\lambda + 3 = (\lambda - 1)(4\lambda - 3) = 0$, giving $\lambda = 1$ or $\lambda = \frac{3}{4}$.

- $\lambda = 1$: the first row gives $\frac{1}{4}A + \frac{3}{4}B = A$, so $A = B$. The ratio is 1, and such a population stays exactly the same each year.
- $\lambda = \frac{3}{4}$: the first row gives $\frac{1}{4}A + \frac{3}{4}B = \frac{3}{4}A$, so $A = \frac{3}{2}B$. The ratio is $\frac{3}{2}$; check the second row: $\frac{3}{2}B - \frac{1}{2} \cdot \frac{3}{2}B = \frac{3}{4}B$. Both species shrink by a factor $\frac{3}{4}$ each year.

Equivalently, without matrices: if $A = rB$ then the new ratio is $\dfrac{\frac{1}{4}r + \frac{3}{4}}{\frac{3}{2} - \frac{1}{2}r}$, and setting this

equal to r gives $2r^2 - 5r + 3 = 0$, so $r = 1$ or $r = \frac{3}{2}$. Either way, the ratio can stay fixed only if it is 1 or $\frac{3}{2}$, as required. We take $x = 1$ and $y = \frac{3}{2}$.

Subsequent years

Write

$$\begin{pmatrix} a \\ b \end{pmatrix} = p \begin{pmatrix} 1 \\ 1 \end{pmatrix} + q \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix}.$$

Comparing components, $p + \frac{3}{2}q = a$ and $p + q = b$. Subtracting, $\frac{1}{2}q = a - b$, so

$$q = 2(a - b), \qquad p = b - q = 3b - 2a.$$

Because M is linear, after one year the first piece is multiplied by 1 and the second by $\frac{3}{4}$; after n years (as long as the model still applies) we have

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = (3b - 2a) \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2(a - b)\left(\frac{3}{4}\right)^n \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix},$$

that is,

$$a_n = (3b - 2a) + 3(a - b)\left(\tfrac{3}{4}\right)^n,$$
$$b_n = (3b - 2a) + 2(a - b)\left(\tfrac{3}{4}\right)^n.$$

Since $(\frac{3}{4})^n \to 0$, the populations head towards $(3b - 2a,\ 3b - 2a)$, a point on the line of equilibria $a = b$.

One reassurance before classifying: the aardvarks never cause trouble. If $a_n \geq 0$ and $b_n > 0$ then $a_{n+1} = \frac{1}{4}a_n + \frac{3}{4}b_n > 0$, so while the buffaloes survive, the aardvarks stay positive too. So the only thing to watch is the sign of b_n, and we work in the region $a \geq 0$, $b > 0$ of the plane.

Ask yourself Could b_n dip below zero for a while and then recover towards a positive limit?

No. In the formula for b_n, only $(\frac{3}{4})^n$ depends on n, and it decreases steadily from 1 to 0. So b_n moves monotonically from $b_0 = b$ towards its limit $3b - 2a$, and every b_n lies between b and $3b - 2a$. That makes the classification clean.

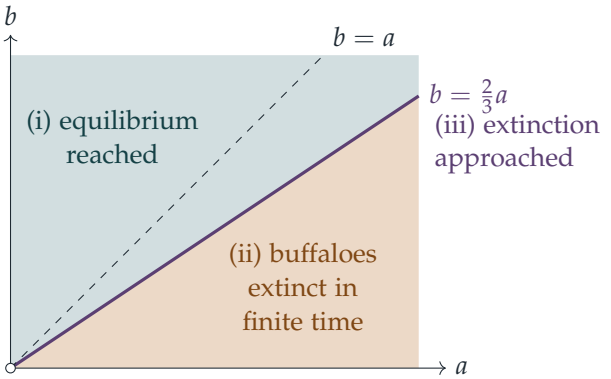
The three regions

(i) If $3b - 2a > 0$, i.e. $b > \frac{2}{3}a$, then b and $3b - 2a$ are both positive, so every b_n is positive and the model applies for ever. The population tends to the equilibrium $(3b - 2a,\ 3b - 2a)$, which is non-zero. This is the region above the line $b = \frac{2}{3}a$ (including the b-axis, $a = 0$).

(ii) If $3b - 2a < 0$, i.e. $b < \frac{2}{3}a$, then $a > \frac{3}{2}b > b$, so $a - b > 0$ and b_n decreases strictly towards the negative number $3b - 2a$. Hence $b_n \leq 0$ for some finite n: the

buffaloes become extinct in finite time. This is the region between the line $b = \frac{2}{3}a$ and the a-axis.

(iii) If $3b - 2a = 0$, i.e. $b = \frac{2}{3}a$ with $a > 0$, then $p = 0$ and the population is a pure multiple of $(\frac{3}{2}, 1)$: $b_n = b\,(\frac{3}{4})^n$, which is positive for every n but tends to 0. The buffaloes (and the aardvarks) approach extinction but never reach it. This is just the half-line $b = \frac{2}{3}a$, $a > 0$.



The dashed line $b = a$ consists of populations that are already in equilibrium; every starting point in region (i) moves towards it, sliding parallel to the line $b = \frac{2}{3}a$ (the direction $(\frac{3}{2}, 1)$ shrinks away while the $(1, 1)$ component stays fixed).

Checks

- With $n = 0$ the formulas give $a_0 = 3b - 2a + 3a - 3b = a$ and $b_0 = 3b - 2a + 2a - 2b = b$, as they must.
- Start at $(a, b) = (1, 3)$, in region (i). One year of the model gives $(\frac{1}{4} + \frac{9}{4},\ \frac{9}{2} - \frac{1}{2}) = (\frac{5}{2}, 4)$, and the formula gives $7 - 6(\frac{3}{4}) = \frac{5}{2}$ and $7 - 4(\frac{3}{4}) = 4$. The limit is $(7, 7)$.
- A computer iterated the model exactly (with fractions) from 3000 random starting points, confirmed the closed form for 40 years, and confirmed that the sign of $3b - 2a$ correctly predicts the outcome in every case, including points on $b = \frac{2}{3}a$, where e.g. $(3, 2)$ becomes $(\frac{729}{1024}, \frac{243}{512})$ after five years, a multiple $(\frac{3}{4})^5$ of the start.

Common slips

- Getting the ratio upside down: the question asks for aardvarks to buffaloes, so the values are 1 and $\frac{3}{2}$, not 1 and $\frac{2}{3}$.
- Forgetting that the model stops when $b_n \leq 0$, and then claiming that every population tends to $(3b - 2a, 3b - 2a)$, including ones with a negative "limit".
- Describing region (iii) as an area: it is only the half-line $b = \frac{2}{3}a$, the boundary between (i) and (ii).

Beyond STEP

This is a discrete linear dynamical system, and the method (diagonalise, then raise each eigenvalue to the nth power) is exactly how first-year linear algebra computes M^n. The eigenvalue 1 gives a line of fixed points and the eigenvalue $\frac{3}{4} < 1$ gives a direction that decays; the same picture, with continuous time and $e^{\lambda t}$ in place of λ^n, governs systems of linear differential equations. Strang's *Introduction to Linear Algebra* has a good chapter on difference equations and powers of matrices, and Boyce and DiPrima's *Elementary Differential Equations* does the continuous version.

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